

A safety claim is only as strong as its comparison.

An independent statistics project that reproduces selected Waymo safety comparisons and shows how geography and assumptions change their interpretation.

THE REFERENCE CHANGES THE RESULT / SF AIRBAG

82.5% to 86.5%

Estimated reduction in Waymo's observed airbag-deployment rate relative to the human reference, before and after geographic matching.

5 events / 16.032 million rider-only miles. The Waymo rate stays at 0.312 events per million miles; the human benchmark moves from 1.787 to 2.313.

Two investigations. Separate historical cohorts.

01 REPRODUCE A PUBLICATION

Can we reproduce the paper's numbers?

For the SF injury reference case, 1 event over 1.755 million rider-only miles yields a rate ratio of 0.0979 against the supplied human benchmark. All four counts match; 12 ratio/endpoint checks differ by less than 0.01 from the publication.

Finding: the appendix code and Equation 2 use different tail probabilities for positive counts. Both conventions are exposed in the app.

02 MATCH GEOGRAPHIC EXPOSURE

Does a fairer reference change the result?

Reconstruct nine published spatial benchmarks from 987 cells across San Francisco, Phoenix and Los Angeles. Every benchmark and all nine event counts match. Sliders reveal sensitivity to mileage weighting and benchmark scale.

For the SF airbag example, the conditional 95% rate-ratio interval is 0.0438-0.3146. It includes only Waymo count uncertainty; the human reference is held fixed.

[Open the project and guided tour: vivaran.news/waymo-project/brief](https://vivaran.news/waymo-project/brief)

How to read the evidence.

The method, the boundaries, and a route to reproduce the result.

COUNT / DISTANCE / REFERENCE

Rate = events / miles. Rate ratio = Waymo rate / human benchmark.

Estimated reduction = $100 \times (1 - \text{rate ratio})$.

Keep these datasets separate

Publication study	arXiv:2312.12675v3; rider-only operations through October 2023. 73 appendix event rows and four reference comparisons. Author-supplied classifications and human benchmark estimates remain dependencies.
Geographic study	March 19, 2025 release; rider-only operations through December 2024, with 2022 human benchmarks. Cells cover 99.67%-99.89% of the three cities' reported miles. Austin has no cell data in this release.
DMV companion	Historical 2020-2024 testing reports. Disengagements are interventions, not crashes. Testing miles are never joined to the rider-only crash cohorts. The experiment planner is hypothetical.

What this does not establish

Geographic matching does not control every difference in driving conditions and does not establish causality. The displayed geographic intervals include only Poisson uncertainty in Waymo counts, excluding uncertainty in human benchmarks, mileage, reporting adjustment and spatial weights. The project does not claim Waymo is safer everywhere or estimate the risk of your next ride.

Inspect, reproduce, challenge

[Publication method and interval-convention audit](#)

[Spatial formula, coverage gaps and source provenance](#)

[Source code, verification runs and local setup](#)

SHA-256 source manifests, preserved event rows, semantic input checks and Python/JavaScript reference checks make the work inspectable. Rebuild the studies using the commands in the linked technical notes. These are historical snapshots, not a live safety ranking.